

GDX Trailing-Return Reversal Strategy

An end-to-end research write-up: data, EDA, model fitting, walk-forward validation, realistic-cost backtesting, robustness checks, and statistical significance testing on gold-miner ETF data

Universe: SPY, IAU, GLD, TLT, GDX, GDXJ daily close prices, 2006-05-22 to 2021-12-31 (3,932 trading days). All model development, tuning, and validation in this report use **train + validation only** (2006-05-22 to 2018-11-15, 3,146 days). The final 20% of the data (2018-11-16 to 2021-12-31, 786 days) is a reserved holdout split that has not been touched anywhere in this report.

Bottom line up front: the research process below found a config that is profitable in-sample after realistic costs, but it does not survive scrutiny -- it fails a statistical significance test, is fragile to small parameter perturbations, and its profits are concentrated in two of four sub-periods rather than persistent. It is frozen (results/FROZEN_CONFIG.json) for methodological discipline, not because it is a validated trading edge.

Executive Summary

- **Data:** six ETFs' daily close prices were parsed from a raw spreadsheet into a clean panel, then split chronologically into train (48%), validation (32%), and a never-touched holdout (20%).
- **EDA** found IAU and GLD are near-duplicate instruments (0.995 correlation), GDX/GDXJ are highly but not perfectly correlated (0.93), GDX's own daily returns show essentially no autocorrelation, and the GDX/GLD relationship is regime-dependent (60-day rolling correlation ranged 0.46-0.92).
- A simple, fully-invested, zero-cost **naive reversal rule** on GDX showed a large paper Sharpe (0.67) and CAGR (21.3%) on train+validation -- explicitly flagged at the time as almost certainly not investable once turnover and costs are considered.
- A causal **feature set** (trailing returns, volatility, RSI, cross-asset momentum) was built with programmatic zero-lookahead verification, and a **logistic regression** fit on the train split alone came back weak: AUC 0.543, 0 of 10 coefficients significant at $p < 0.05$.
- A **walk-forward grid search** (16 lookback x threshold combos, quarterly-refit direction, no costs) found a best Sharpe of 0.379 -- but re-running the top 5 through a **realistic-cost backtest** (real commission/slippage, fixed direction, persistent positions) left only 1 of 5 profitable, with a CAGR of just +0.44%.
- **Robustness checks** on that one surviving config found: only 2 of 9 nearby parameter combinations were profitable, only 2 of 4 equal sub-periods were profitable, and doubling costs flipped CAGR to -13.8%.
- A **t-test** found the mean daily strategy return is not statistically distinguishable from zero ($p=0.578$), and a **bootstrap** of trade-level Sharpe ratios produced a 5th-95th percentile interval of [-0.33, 0.64] -- straddling zero.
- The final parameters were **frozen** (results/FROZEN_CONFIG.json) with these limitations recorded verbatim, so a holdout evaluation -- if one is ever run -- uses an untouched, honestly-labeled config.

1. Data and Chronological Splits

The source file (data/raw/gdx_data.xlsx) contained a single sheet with no assumed structure going in -- the header row, date column, and per-ticker missing-data patterns were discovered programmatically before anything else was built. GDXJ has no data before 2009-11-11 (it hadn't launched); every downstream step treats those cells as missing, not zero.

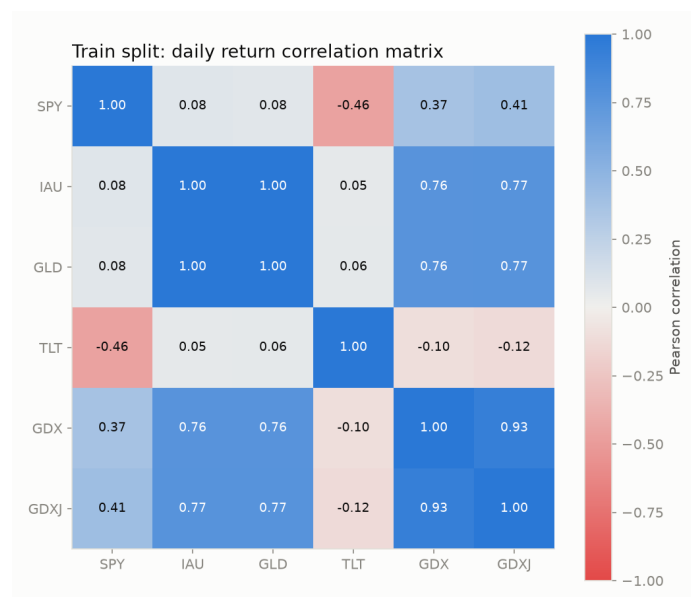
The resulting panel (3,932 rows) was split once, chronologically, by row count -- holdout carved out first as the most recent 20%, then the remaining 80% split 60/40 into train and validation:

Split	Date range	Rows	Share
Train	2006-05-22 to 2013-11-18	1,888	48%
Validation	2013-11-19 to 2018-11-15	1,258	32%
Holdout (untouched)	2018-11-16 to 2021-12-31	786	20%

The holdout file is literally named `DO_NOT_TOUCH_holdout.parquet` and has not been loaded, inspected, or fit on anywhere in this project.

2. Exploratory Data Analysis (Train Split Only)

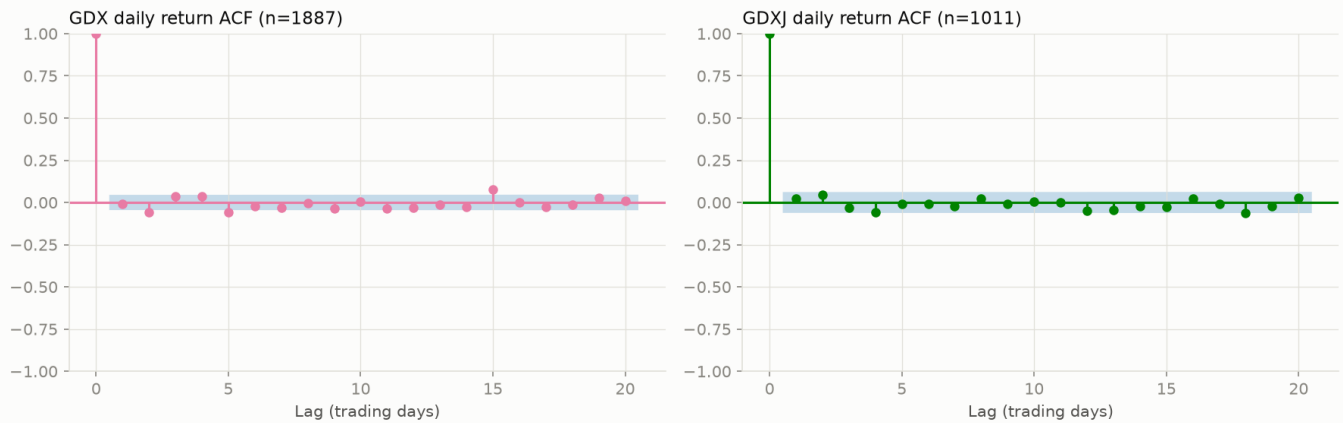
Daily-return correlation matrix and autocorrelation of GDX/GDXJ returns, computed on the train split only:



Daily return correlation matrix (train split).

- **IAU vs. GLD: 0.995** -- near-duplicate instruments; using both as independent model features risks severe multicollinearity.
- **GDX vs. GDXJ: 0.93** -- highly related but not interchangeable; GDXJ is the higher-beta junior-miner cousin.
- **Gold ETFs vs. GDX/GDXJ: ~0.76** -- miners track gold but roughly 40% of return variance is unexplained by gold alone.
- **SPY vs. TLT: -0.46** -- the expected stocks/bonds hedge relationship.

Train split: autocorrelation of daily returns, 20 lags (shaded = 95% CI)



Autocorrelation of GDX and GDXJ daily returns, 20 lags (shaded = 95% CI). Both are close to serially uncorrelated -- no obvious short-horizon momentum or mean-reversion signal from lagged returns alone.

A 60-day rolling GDX/GLD correlation ranged from 0.46 to 0.92 (mean 0.79) -- weakening sharply during fast, large gold moves (e.g. the August 2011 rally), which is exactly when a fixed-correlation assumption would be most misleading. See `results/eda_summary.md` for the full write-up.

3. Naive Reversal Baseline (No Costs) and Timing Convention

Before any modeling, an explicit timing specification was written down (entry at close $t+1$, exit at close $t+2$, relative to a signal known at close t) precisely because the raw data is close-price-only -- there is no open/intraday price to enter at, so this was chosen as the earliest honest execution point and verified programmatically to have zero lookahead (signal recomputed independently from raw prices and checked to match; entry/exit dates checked to land strictly later than the signal date).

Under that timing convention, a simple, fully-invested, **zero-cost** daily reversal rule ($-\text{sign}(\text{prior-day return})$) was compared against buy-and-hold on train+validation:

Strategy	Total return	CAGR	Ann. vol	Sharpe	Max DD
Buy-and-hold GDX	-44.6%	-4.6%	42.4%	0.10	-80.5%
Buy-and-hold SPY	178.6%	8.6%	19.0%	0.53	-55.2%
Naive reversal on GDX	1,007.1%	21.3%	42.3%	0.67	-68.2%

This was flagged at the time as almost certainly not investable as shown: it trades every single day (100% daily turnover) with zero transaction costs, slippage, or borrow cost assumed. Everything from Section 6 onward exists to find out how much of this apparent edge survives realistic execution.

4. Causal Features and In-Sample Model Fit

A feature set was built from train+validation data using strictly causal rolling windows (GDX trailing returns at 5/10/20/60 days, 20-day realized volatility, RSI(14), GLD/IAU/TLT 20-day momentum, GDX/GDXJ relative momentum), with the label defined as next-day GDX return. Zero-lookahead was proven programmatically via **truncation invariance**: recomputing every feature on data cut off at any date reproduces identical values up to that date. This check was confirmed to actually catch bugs by injecting a deliberate centered-rolling-window error and watching it fail; 17 automated tests (`tests/test_no_leakage.py`) cover this.

Fit on the **train split only**: a threshold rule on trailing return, and a logistic regression on all ten features predicting next-day direction.

Model	Result
Threshold rule (best of 4 windows)	52.9% accuracy vs. 50.8% majority baseline (+2.1pp)
Logistic regression	AUC 0.543, train accuracy 53.2% vs. 50.9% baseline
Significant coefficients	0 of 10, at $p < 0.05$

This in-sample fit was already weak and consistent with the EDA's near-zero autocorrelation finding -- GDX's own past return carried little signal for its next-day direction, even before any holdout or cost considerations.

5. Walk-Forward Grid Search (No Costs, Adaptive Direction)

A quarterly-refit, expanding-window walk-forward evaluation was run on train+validation across a grid of lookback (5/10/20/60 days) x threshold (0.5%/1%/1.5%/2%) -- 16 combinations. Direction (momentum vs. reversal) was re-picked every quarter based on which had the higher realized Sharpe in the expanding window so far; no transaction costs were modeled. Top 5 by walk-forward Sharpe:

Rank	Lookback	Threshold	Walk-fwd Sharpe	Cumulative return	Max DD
1	20d	2.0%	0.379	+139.3%	-68.3%
2	60d	0.5%	0.251	+22.4%	-69.3%
3	60d	2.0%	0.232	+14.4%	-66.7%
4	60d	1.5%	0.200	-2.9%	-75.4%
5	60d	1.0%	0.197	-5.5%	-71.6%

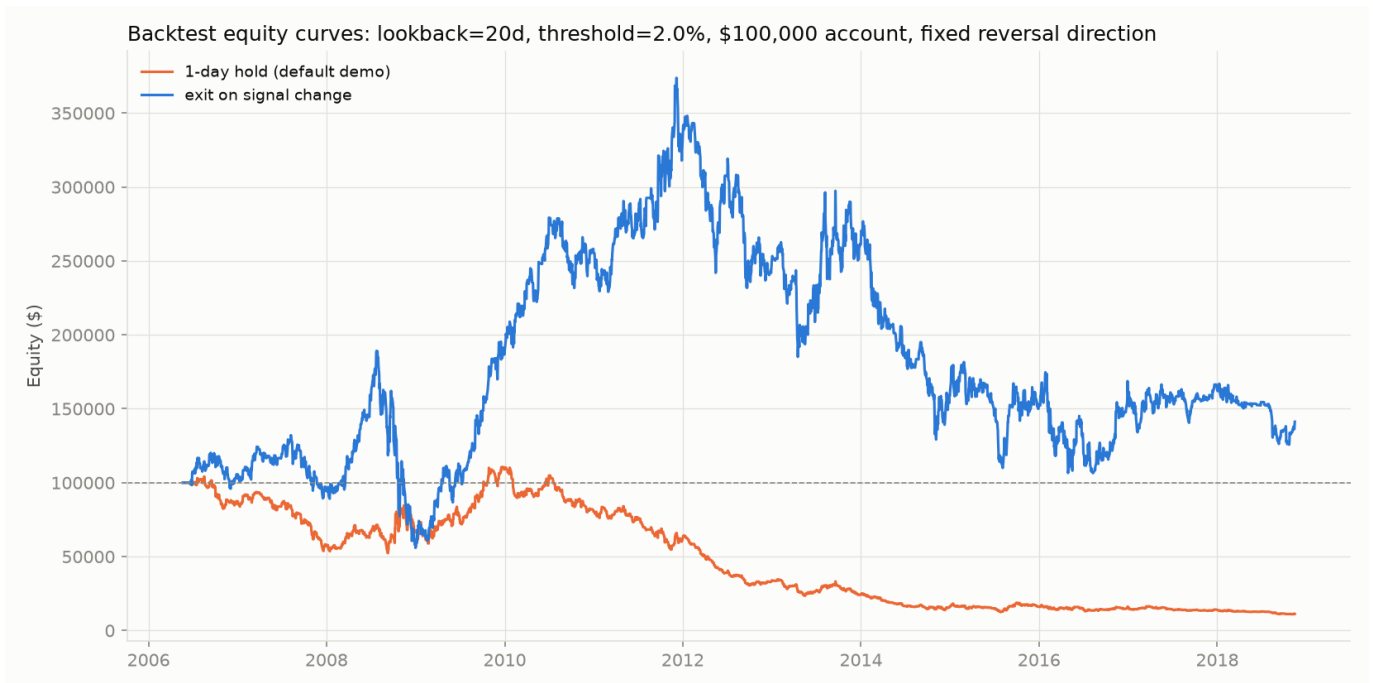
This grid's optimism turned out to hinge heavily on two things it didn't model: zero transaction costs, and an adaptive direction that a real deployment would need to replicate somehow. Section 6 tests exactly that.

6. Realistic-Cost Backtest of the Top 5 Configs

A custom backtest engine (market-on-close execution, configurable commission + slippage, % -of-equity position sizing with integer share lots, pluggable exit rules) re-ran the top 5 walk-forward configs with **\$0.005/share commission + 0.05% slippage per leg**, a \$100,000 account, **fixed** reversal direction (not the adaptive quarterly-refit direction above), and a 1-day hold:

Rank	Lookback / Threshold	Walk-fwd Sharpe	Backtest CAGR	Profitable after costs?
1	20d / 2.0%	0.379	+0.44%	Yes (\$5,618 net)
2	60d / 0.5%	0.251	-18.2%	No (-\$91,866)
3	60d / 2.0%	0.232	-8.0%	No (-\$64,818)
4	60d / 1.5%	0.200	-16.4%	No (-\$89,282)
5	60d / 1.0%	0.197	-15.5%	No (-\$87,833)

Only rank 1 survived, and the reason is **not mainly transaction costs**: rank 1's gross P&L; at quoted prices was ~\$233K against only ~\$59K of commission. The dominant effect is that the walk-forward grid earned its Sharpes under an **adaptive**, quarterly-refit direction, while this backtest uses one **fixed** direction for the whole 12.5-year period -- being on the wrong side of a volatile, trending asset for extended stretches at 100% position sizing is costly.

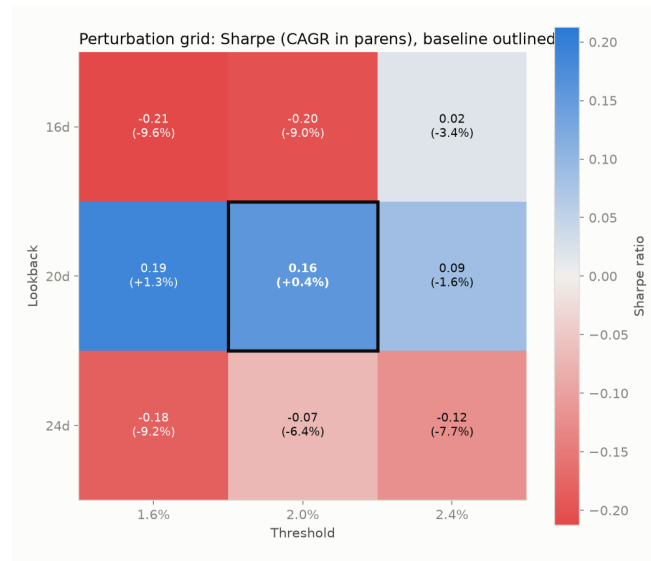


Rank-1 config: fixed 1-day hold (orange) vs. an alternative exit rule, exit-on-signal-change (blue). Same signal, same costs -- exit rule alone swings the outcome from a loss to a substantial gain.

7. Robustness Checks on the Single Surviving Config

The only profitable-after-costs config (lookback=20d, threshold=2.0%) was stress-tested four ways, all on train+validation with the same costs/exit rule/direction unless a check explicitly varied one.

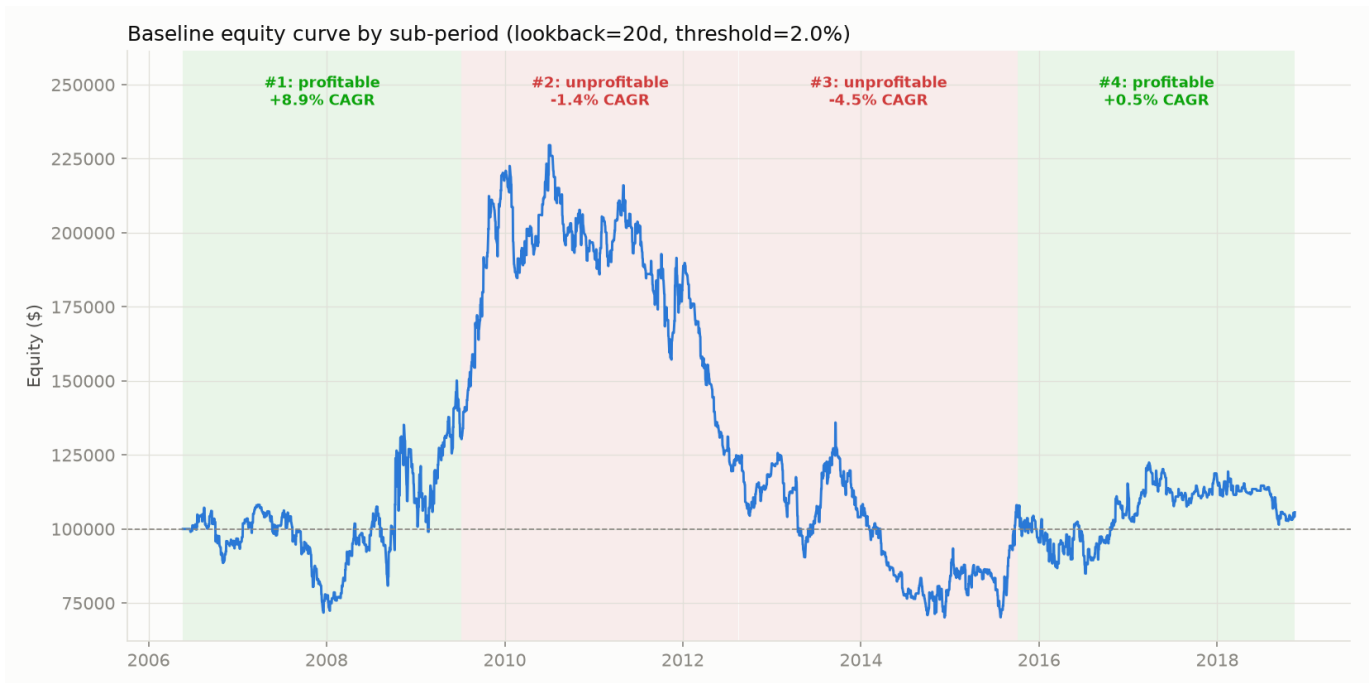
7.1 Parameter perturbation (+/-20% on lookback and threshold)



Sharpe across the 3x3 grid (CAGR in parens); baseline outlined. Only 2 of 9 nearby configs are profitable.

A config whose immediate neighbors in parameter space are mostly unprofitable is a narrow, not-very-robust local optimum rather than a broad, stable edge.

7.2 Performance by sub-period (4 equal chronological chunks)

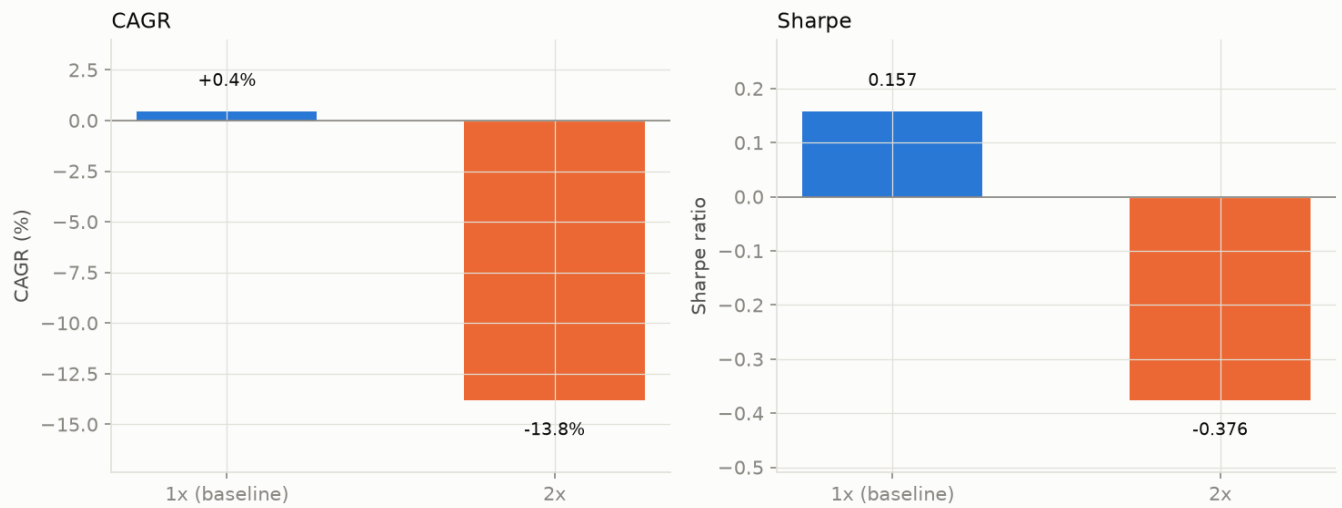


Baseline equity curve, sliced into 4 equal sub-periods and shaded by profitable (green) / unprofitable (red). Only 2 of 4 are profitable.

Sub-period	Dates	CAGR	Sharpe	Profitable?
1	2006-05-22 to 2009-07-07	+8.9%	0.413	Yes
2	2009-07-08 to 2012-08-17	-1.4%	0.052	No
3	2012-08-20 to 2015-10-05	-4.5%	-0.011	No
4	2015-10-06 to 2018-11-15	+0.5%	0.139	Yes

7.3 Cost sensitivity (2x commission and slippage)

Cost sensitivity: 1x vs. 2x commission and slippage



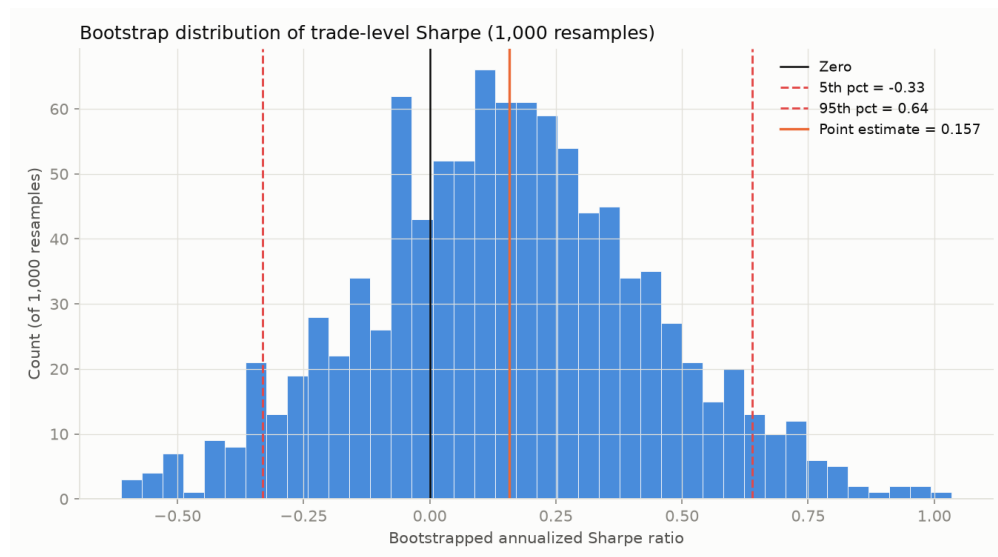
Doubling costs flips CAGR from +0.44% to -13.8% and Sharpe from 0.157 to -0.376.

7.4 Trade frequency: all 4 sub-periods pass a minimum ≥ 1 trade/month bar comfortably (8.7-9.2/month each)
-- trade frequency is not the weak point here.

8. Statistical Significance

Two checks on the same final config, on train+validation:

- **T-test** of mean daily strategy return vs. zero: $t = 0.556$, two-sided $p = 0.578$ -- **not significant** at the 5% level.
- **Bootstrap** of the 1,384 trade-level returns (1,000 resamples, seed=42), each iteration's Sharpe annualized by $\sqrt{\text{trades}/\text{year}}$: point-estimate Sharpe 0.157, but the 5th/95th percentile interval is **[-0.33, 0.64]** -- straddling zero.



Bootstrap distribution of trade-level Sharpe (1,000 resamples). Zero sits well inside the bulk of the distribution, not in a tail.

Both findings restate, in standard statistical form, what the walk-forward optimism, the thin backtest margin, and the robustness fragility already suggested: this is not a validated edge.

9. Frozen Final Configuration

The exact parameters selected from this in-sample process were written to `results/FROZEN_CONFIG.json`, with instructions not to modify or re-tune them further:

Parameter	Value
Ticker	GDX
Signal	Reversal on 20-day trailing return, 2.0% deadzone threshold
Direction	Fixed reversal (not adaptively refit)
Exit rule	1-day hold (<code>exit_after_n_days(0)</code>) -- market-on-close, entry $t+1$ / exit $t+2$
Position sizing	100% of equity, integer share lots
Cost assumptions	\$0.005/share commission + 0.05% slippage per leg
Reference account size	\$100,000 (in-sample evaluation only, not a deployment constraint)

10. Conclusion and Recommendations

Every stage of this research process pointed the same direction: an initially promising signal (a zero-cost naive reversal rule, then a walk-forward grid search) shrank steadily as more realism was added -- real costs, a fixed rather than adaptive direction, parameter neighborhoods, sub-period consistency, and finally formal significance testing. What's left is a config with a CAGR of well under 1%, a Sharpe not distinguishable from zero, and profits concentrated in two specific multi-year windows rather than persistent.

Recommendations:

- Do not treat the frozen config as a trading strategy ready for capital -- treat it as the end product of a disciplined process that happens to have found a thin, fragile, statistically insignificant result.
- If a holdout evaluation is run, it should be a single, final look (per the reserved 786-day split) -- not a new round of tuning against holdout performance, which would undo the entire point of having kept it untouched.
- The adaptive quarterly direction-refit mechanism (walk-forward) meaningfully outperformed the fixed-direction deployment in-sample; if this line of research continues, building that adaptive mechanism into the live/holdout-facing engine (rather than fixing direction once) is the most promising unexplored lever.
- IAU/GLD collinearity should be resolved (drop one, or use a spread/ratio feature) before any further feature-based modeling.

Full detail, code, and every intermediate artifact referenced above are in the project repository under `results/` and `src/`: `eda_summary.md`, `baselines.md`, `timing_spec.md`, `naive_baseline/`, `naive_models/`, `walk_forward_grid.csv`, `backtest_results.csv`, `backtest_results_notes.md`, `robustness_report.md`, `significance_report.md`, `FROZEN_CONFIG.json`.